

# Loops

# Loops

- Branching has allowed us to make various decisions in our code
- What if we wanted to perform an action multiple times?
  - How would we do that now?
    - Copy and paste?
    - Functions (we haven't covered these) and then copy and paste?

# Loops

- Loops allow us to repeat code until a given condition has been met.
- Loops are made of two parts
  - loop expression
  - loop body
- Iteration
  - each time we move through the statements in the loop body is known as an iteration

# Loops

- There are two kinds of Loops
  - While Loops
  - For (each) Loops

# While Loops

- **while loops** runs a block of code repeatedly as long as a the loop's condition is True
  - if it is False the loop ends and does not run the loop's body

# While Loops

- *Sentinel* value is a value that causes a loop to end.
  - can be, but not limited to:
    - number value
    - string value
    - user input

# While Loops

- What good is a while loop?
  - used when you do not know how many times a loop will actually run
- Why wouldn't you use a while loop?
  - Infinite loops
    - A loop never stops running because the condition is always True